

Robustness of Diffusion and Autoregressive Language Models to CheckList-Style Perturbations

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Abstract

We compare diffusion-based and autoregressive language models under CheckList-style perturbations. We frame semantics-preserving perturbations as potentially containing an off-manifold component and hypothesise that diffusion denoising can reduce the resulting output change relative to autoregressive decoding. We evaluate eight instruction-tuned 7–9B models (three diffusion, five autoregressive) across sentiment, QQP, and SQuAD using behavioural pass rates and a perturbation coefficient that measures output change relative to input change. Results are task-dependent. Behaviourally, diffusion gains 18.2 and 11.0 points overall on QQP and SQuAD, while autoregressive models perform better on the sentiment and SQuAD invariance slices. On QQP, the all-model coefficient ratio does not favour diffusion ($\bar{\gamma}_{\text{AR}}/\bar{\alpha}_{\text{Diff}} \approx 0.91$); excluding two autoregressive models with strong label-prior effects as a diagnostic sensitivity analysis yields 1.23 overall and 1.48 on invariance tests. Diffusion also performs worse on perturbations that should change the output. Thus, the observed stability differences are task-conditional, and because the empirical diffusion coefficients exceed one, the experiments do not directly validate the contraction assumption of our theoretical proposition. Data and code are available at: https://github.com/AarushRathore1729/dlm_ar_final.

1 Introduction

Natural language has been found in practice to exhibit approximately heavy-tailed (Zipfian) distributions over tokens and sequences (Zipf, 1949; Manning and Schütze, 1999; Newman, 2005). That is, there is a substantial probability mass associated with rare, but valid, language sequences. For instance, consider the following sentences: (1) “The cat sat on the mat.”, (2) “The feline sat on the mat.”, and (3) “Upon the mat the feline sat.” Readers will agree that all are valid English sentences and

convey the same meaning. However, they need not occur with the same frequency in natural language. It is plausible that the first sentence occurs more frequently than the second, which in turn is more common than the third.

For any language $\mathcal{L}$ consisting of sentences $x \in \mathcal{X}$, we can assume there exists a distribution $P_{\mathcal{L}}$ that is non-zero over valid, meaningful sentences in $\mathcal{L}$. For models trained on sentences from $\mathcal{L}$, we can define a *data manifold* as the subset of inputs whose probability under the training data distribution is sufficiently high. Specifically, if P_{train} denotes the training distribution over the space of textual inputs $\mathcal{X}$, then for a fixed threshold $\tau > 0$, we can define the data manifold as the set

$$\mathcal{M}_{\tau} = \{x \in \mathcal{X} : P_{\text{train}}(x) \geq \tau\}$$

for some small $\tau > 0$. In practice, both $P_{\mathcal{L}}$ and P_{train} are not observable, and we resort to model-based approximations. For the present, it is helpful to continue with this abstraction, and say an instance $x \in \mathcal{X}$ is “off-manifold” if $x \notin \mathcal{M}_{\tau}$. Thus, if Sentence 3 above had a sufficiently low probability, then we would consider it off-manifold. This paper examines how autoregressive (AR) and diffusion-based language models respond to controlled perturbations that we model as potentially moving inputs away from high-density regions. Our conjecture is that the sequential conditioning mechanism in AR models does not necessarily restore such inputs toward high-density regions, whereas diffusion models may benefit from their denoising process. We do not directly measure distance from the data manifold; instead, we study behavioural robustness and perturbation-based output-change measures.

In this work, we use behavioural testing with CheckList (Ribeiro et al., 2020) to compare the recent family of discrete diffusion language models (Austin et al., 2021; Lou et al., 2024; Sahoo et al.,

2024; Nie et al., 2025; Ye et al., 2025) against autoregressive models (Vaswani et al., 2017; Radford et al., 2019). We ask whether the iterative denoising in diffusion models gives them more robustness to meaning-preserving perturbations than autoregressive decoding (Vaswani et al., 2017; Radford et al., 2019), given the known sensitivity of NLP models to distribution-shifting inputs (Jia and Liang, 2017).

2 A Simple Framework for Robustness

We analyse the robustness of language models under CheckList-style perturbations. We characterise such perturbations relative to a high-density region of the training distribution, and ask how the two generation mechanisms respond when an input leaves it. All definitions and assumptions needed for the comparison are stated here.

Definition 1 (Data Manifold). *Let $\mathcal{X}$ denote the space of all token sequences, and let P_{train} denote the (unknown) training distribution over $\mathcal{X}$. For a fixed threshold $\tau > 0$, we define the data manifold as $\mathcal{M}_\tau = \{x \in \mathcal{X} : P_{\text{train}}(x) \geq \tau\}$.*

The set $\mathcal{M}_\tau$ captures the high-density region of the training distribution. $\mathcal{M}_\tau$ should be interpreted as an approximate data manifold for several reasons. First, it is defined via a threshold on the (unknown) training distribution P_{train} , which itself is only an approximation to the underlying language distribution $P_{\mathcal{L}}$. Second, the choice of threshold τ induces an arbitrary boundary on a continuous distribution. Finally, the notion of a manifold serves as a conceptual simplification of a discrete and highly irregular probability landscape. In practice, P_{train} is not directly accessible, and the use of models introduces a further layer of approximation.

Two inputs are *semantically equivalent* ($\text{Sem}(x) \simeq \text{Sem}(x')$) when an abstract semantic map $\text{Sem} : \mathcal{X} \rightarrow \mathcal{S}$ assigns them equivalent representations in a semantic space $\mathcal{S}$; for example, $\mathcal{S}$ may be a space of logical formulae with $\simeq$ denoting logical equivalence, though the specific representation is not essential.

Definition 2 (Semantics-preserving Perturbation). *A semantics-preserving perturbation is a transformation $T : \mathcal{X} \rightarrow \mathcal{X}$ such that, for an input $x \in \mathcal{X}$, the perturbed input $x' = T(x)$ is semantically equivalent to x and $P_{\text{train}}(x) \neq P_{\text{train}}(x')$.*

That is, the perturbations preserve the semantic properties but alter the distributional properties under P_{train} . Our interest is further in perturbations

that can be decomposed into two components, eventually resulting in a lower-probability instance:

Definition 3 (Decomposable Perturbation). *Let $x \in \mathcal{M}_\tau$ and let $x' = T(x)$ be a semantics-preserving perturbation of x . We say T is a decomposable semantics-preserving perturbation, or simply a decomposable perturbation, if x' admits a decomposition into an on-manifold and an off-manifold component; that is, there exists an intermediate point $\tilde{x} \in \mathcal{X}$ such that:*

1. $\text{Sem}(\tilde{x}) \simeq \text{Sem}(x)$ and $\tilde{x} \in \mathcal{M}_\tau$;
2. x' is obtained from $\tilde{x}$ by a further transformation that reduces its likelihood under P_{train} , i.e., $P_{\text{train}}(x') < P_{\text{train}}(\tilde{x})$.

In this case, we interpret the transformation $x \rightarrow \tilde{x}$ as the on-manifold component and $\tilde{x} \rightarrow x'$ as the off-manifold component.

Remark 1. *For notational convenience, we may refer to the transformation $x \rightarrow \tilde{x}$ as the $\epsilon_{\parallel}$ component and $\tilde{x} \rightarrow x'$ as the $\epsilon_{\perp}$ component. This decomposition is conceptual, reflecting the distinction between semantic invariance and distributional shift, rather than a literal vector decomposition in $\mathcal{X}$.*

We will assume CheckList-style perturbations can be formalised as a class of decomposable semantics-preserving perturbations.

Definition 4 (CheckList-style perturbation family). *Let $\mathcal{T}$ be a collection of transformations $T : \mathcal{X} \rightarrow \mathcal{X}$. We say that $\mathcal{T}$ defines a CheckList-style perturbation family if the following conditions hold:*

1. *Each $T \in \mathcal{T}$ is a semantics-preserving transformation;*
2. *Each $T \in \mathcal{T}$ is a local transformation. That is, it modifies x in a controlled and limited manner (for example, lexical substitution, syntactic reordering), rather than arbitrarily rewriting the entire sequence.*
3. *Each $T \in \mathcal{T}$ corresponds to identifiable perturbation types (for example synonym substitution, negation, reordering), and can be applied consistently across inputs.*

We are now able to meaningfully assign a distribution over the transformations in a CheckList family.

Definition 5 (Perturbation distribution). *A perturbation distribution is a probability distribution $\mathcal{D}$ over a family of CheckList-style transformations $\mathcal{T}$. A perturbation is generated by first sampling $T \sim \mathcal{D}$ and then applying T to an input x .*

As defined, nothing requires perturbations to result in off-manifold instances. We introduce this requirement as an assumption.

Assumption 1 (off-manifold perturbation distribution). *Let $\mathcal{T}$ be a collection of transformations and let $\mathcal{D}$ be a distribution over $\mathcal{T}$. We assume that $\mathcal{D}$ assigns non-zero probability to transformations that produce inputs outside $\mathcal{M}_\tau$.*

In this paper, we assume we will only be dealing with perturbations that satisfy the off-manifold assumption. That is, with non-zero probability over $x \sim P_{\text{train}}$ and $T \sim \mathcal{D}$, the perturbation $T(x)$ admits a decomposition with $\epsilon_\perp \neq 0$ in the sense described in Remark 1.

Comparative Robustness. For a model f and discrepancy L , the expected degradation is $\Delta(f) = \mathbb{E}_{x,T}[L(f(T(x)), f(x))]$. We compare models by their expected off-manifold degradation $\Delta_\perp(f) = \mathbb{E}[L(f(T(x)), f(x)) \mid T(x) \notin \mathcal{M}_\tau]$: a model f is more robust than g under $\mathcal{D}$ if $\Delta_\perp(f) < \Delta_\perp(g)$.

2.1 Application to Autoregressive and Diffusion Models

Autoregressive (AR) language models generate outputs sequentially, with each prediction conditioned on previously generated tokens. Diffusion-based models (Diff), on the other hand, generate outputs through an iterative denoising process. This broad difference suggests the following conjecture:

Conjecture. Under perturbations that introduce a non-trivial off-manifold component ($\epsilon_\perp$), diffusion-based language models exhibit lower expected degradation in performance, and hence are more robust than autoregressive models.

We note that a slightly weaker form of this readily follows under some assumptions about language models f_{AR} and f_{Diff} , and the loss function L . In the following we assume a discrepancy measure $d(\cdot, \cdot)$ on pairs from $\mathcal{X}$.

Assumption 2 (Autoregressive-amplification). *There exists $\gamma > 1$ such that, for inputs with a non-trivial off-manifold component,*

$$d(f_{\text{AR}}(T(x)), f_{\text{AR}}(x)) \geq \gamma d(T(x), x).$$

Assumption 3 (Diffusion-contraction). *There exists $\alpha \in (0, 1)$ such that*

$$d(f_{\text{Diff}}(T(x)), f_{\text{Diff}}(x)) \leq \alpha d(T(x), x).$$

Assumption 4 (Scaled-loss function). *There exist constants $c, C > 0$ such that for all $u, v \in \mathcal{X}$,*

$$c d(u, v) \leq L(u, v) \leq C d(u, v).$$

Proposition 1. *Let f_{AR} and f_{Diff} denote autoregressive and diffusion-based language models, respectively. Let $\mathcal{D}$ be a perturbation distribution. Let (A1) $\mathcal{D}$ be an off-manifold perturbation; (A2) f_{AR} satisfy the Autoregressive-amplification assumption with amplification factor γ ; (A3) f_{Diff} satisfy the Diffusion-contraction assumption with contraction factor α ; and (A4) degradation calculations use a scaled-loss function with constants c, C .*

If

$$\frac{C}{c} \cdot \frac{\alpha}{\gamma} < 1$$

then

$$\Delta_\perp(f_{\text{Diff}}) < \Delta_\perp(f_{\text{AR}}).$$

The proof is deferred to Appendix A. The result is subject to $\frac{C}{c} \cdot \frac{\alpha}{\gamma} < 1$. We can interpret this constraint as requiring the diffusion-side multiplicative bound to be sufficiently smaller than the AR-side bound, relative to the constants induced by the loss. Next, we examine the extent to which these effects can be translated into practice, which requires replacing theoretical quantities with observable (or measurable) counterparts.

3 Empirical Evaluation

3.1 Aim

We test whether, under the evaluated CheckList-style perturbations, diffusion-based language models exhibit lower output-change and/or higher behavioural robustness than autoregressive models. Within our theoretical framing, we model some of these perturbations as having a non-trivial off-manifold component; this condition is an assumption and is not measured for individual perturbation pairs.

3.2 Materials

Models. We evaluate eight instruction-tuned language models in the 7–9B range. The three *diffusion* models (DLMs) are LLaDA-8B-Instruct (Nie et al., 2025), LLaDA-MoE-7B-A1B-Instruct (Zhu

et al., 2025), and Dream-v0-7B-Instruct (Ye et al., 2025). The five *autoregressive* models (ARs) are Llama-3.1-8B-Instruct (Llama Team, AI @ Meta, 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Qwen2.5-7B-Instruct (Yang et al., 2024), Gemma-2-9B-Instruct (Gemma Team, 2024), and OLMo-2-7B-Instruct (Team OLMo et al., 2024). We treat the three DLMs and the five ARs as two *families* and report family-level comparisons alongside per-model values, so that no single model drives a conclusion.

Suites. We use the three original CheckList behavioural suites: **sentiment** (sentiment analysis), **QQP** (duplicate-question detection), and **SQuAD** (extractive QA). Each suite is organised into three test families: **MFT** (Minimum Functionality, a baseline input/output correctness check), **INV** (Invariance, the output should *not* change under a semantics-preserving perturbation such as a paraphrase, typo, or entity swap; this is the closest empirical analogue to the semantics-preserving setting of Proposition 1), and **DIR** (Directional, the output *should* change in a specified direction; the sensitivity probe). Within each family the perturbations are grouped into *capabilities* (e.g. NER, negation, vocabulary, fairness), which we use to break down where the diffusion stability advantage holds. These three suites admit *both* a pass rate *and* the perturbation-coefficient ratio, so all empirical comparisons use the same behavioural and coefficient-based measurements.

3.3 Method

Framework and hypothesis. This section estimates empirical analogues of the framework’s quantities. A perturbation T maps $x \rightarrow x' = T(x)$; for the theoretical framework, we model the relevant perturbations as semantics-preserving and as carrying a non-trivial off-manifold component. These conditions are not directly verified per pair.

Estimator. For each perturbation pair (x, x') we compute the per-pair perturbation coefficient

$$\rho(f) = \frac{d(f(x'), f(x))}{d(x', x)},$$

writing $\gamma_{\text{AR}} = \rho(f_{\text{AR}})$ for AR models and $\alpha_{\text{Diff}} = \rho(f_{\text{Diff}})$ for DLMs, and aggregate to pair-weighted means $\bar{\gamma}_{\text{AR}}, \bar{\alpha}_{\text{Diff}}$ over the holdout. The quantity under test is the ratio $\gamma/\alpha = \bar{\gamma}_{\text{AR}}/\bar{\alpha}_{\text{Diff}}$: a value > 1 means the diffusion model amplifies a meaning-preserving perturbation *less* than the AR

Input: AR models $\mathcal{F}_{\text{AR}}$, diffusion models $\mathcal{F}_{\text{Diff}}$; CheckList INV/DIR suites.

Output: $\bar{\gamma}_{\text{AR}}, \bar{\alpha}_{\text{Diff}}, \bar{\gamma}_{\text{AR}}/\bar{\alpha}_{\text{Diff}}$; diffusion has a lower perturbation coefficient on a suite iff $\bar{\alpha}_{\text{Diff}} < \bar{\gamma}_{\text{AR}}$.

Suite-specific output discrepancy, $d_s(y', y)$:

Suite	$d_s(y', y)$	type
Sentiment	ordinal dist. {neg, neu, pos}	3-level
QQP	$\mathbf{1}[y \neq y']$	binary
SQuAD	$1 - \cos(\text{emb}(y), \text{emb}(y'))$	free-text

Input side $d(x', x) = 1 - \cos(\text{emb}(x), \text{emb}(x'))$, shared.

- 1: $R[f] \leftarrow \emptyset$ for every model f
- 2: **for** each suite s , INV/DIR subtest **do**
- 3: **for** each base input x **do**
- 4: $T \sim \mathcal{D}$; $x' \leftarrow T(x)$
- 5: **for** each model f **do**
- 6: $y \leftarrow f(x), y' \leftarrow f(x')$
- 7: $\rho \leftarrow d_s(y', y) / d(x', x)$
- 8: append ρ to $R[f]$
- 9: $\bar{\alpha}_{\text{Diff}} \leftarrow$ pair-wt. mean over $\mathcal{F}_{\text{Diff}}$
- 10: $\bar{\gamma}_{\text{AR}} \leftarrow$ pair-wt. mean over all $\mathcal{F}_{\text{AR}}$; report label-prior-excluded estimate separately
- 11: attach 95% bootstrap CIs (10^4 resamples)
- 12: **return** $\bar{\gamma}_{\text{AR}}, \bar{\alpha}_{\text{Diff}}, \bar{\gamma}_{\text{AR}}/\bar{\alpha}_{\text{Diff}}$

Procedure 1: Estimation protocol for the empirical perturbation-coefficient comparison related to Proposition 1. The per-pair factor ρ uses a suite-specific output discrepancy d_s , reflecting the different label spaces of the three CheckList tasks.

baseline. We use the relative ordering $\bar{\alpha}_{\text{Diff}} < \bar{\gamma}_{\text{AR}}$ (equivalently $\gamma/\alpha > 1$) as an empirical diagnostic. This ordering alone does not establish the contraction and amplification assumptions of Proposition 1.

Output-change metrics. Because the right notion of “output change” is task-dependent, we measure $d(f(x'), f(x))$ three ways and treat agreement across them as evidence of a robust finding. (i) *Task-aware (primary)*: the suite-specific metrics d_s listed in Procedure 1. (ii) *Output-embedding companion*: the same estimator with $d = 1 - \cos(\text{emb}(y), \text{emb}(y'))$ for *all* tasks, to check that the conclusion does not depend on label-space choices. (iii) *Embedding distance*: $1 - \cos(\text{emb}(y), \text{emb}(y'))$ on sentence embeddings (all-MiniLM-L6-v2), aggregated per subtest.

Protocol. Procedure 1 gives the full estimator. The CheckList INV/DIR templates supply controlled perturbation pairs (lines 2–4). INV provides the closest empirical analogue to the semantics-preserving setting of Proposition 1, while DIR serves as a sensitivity probe. Off-manifoldness is a modelling assumption rather than a quantity

Suite	Overall gap (pp)	INV-only gap (pp)
Sentiment	+1.6 [−2.2, +5.7]	−5.8 [−11.0, −1.6]
QQP	+18.2 [+5.0, +32.4]	−0.8 [−1.8, +0.3]
SQuAD	+11.0 [+3.1, +19.4]	−6.8 [−9.4, −4.2]

Table 1: Diffusion-family minus AR-family macro pass-rate gap (percentage points) with 95% bootstrap CIs over subtests. **Bold**: diffusion family significantly higher; underlined: AR family significantly higher.

measured per pair. We decode $y = f(x)$ and $y' = f(x')$ for every model (lines 5–6), compute the input- and output-side discrepancies and the per-pair factor ρ (line 7), and aggregate to pair-weighted means (lines 9–10). We also report the conditional degradation $\hat{\Delta}_{\perp}$ as family pass rates (§4.1) and the comparison $\bar{\alpha} < \bar{\gamma}$ (line 12). We attach 95% bootstrap confidence intervals (line 11; 10,000 resamples) to every aggregate: a paired resample over common subtests for ratio gaps, and a subtest (or, for fewer than five subtests, case-level binomial) resample for pass-rate gaps. A subtest-level concordance check between the ratio and the behavioural fail rate (96% agreement on INV, 100% on DIR; MFT diverges by design, since it rewards correctness rather than stability) scopes the ratio estimator to the INV/DIR families. For the DLMs we impose a decoding-budget floor of $2\times$ the maximum new tokens, which removes an under-denoising artefact (e.g. Dream SQuAD invalid spans fall $25\% \rightarrow 3\%$).

4 Results

4.1 Layer 1: behavioural pass rates

Table 1 reports the diffusion-family minus AR-family macro pass rate per suite, overall and INV-only, with 95% bootstrap CIs. The diffusion family wins significantly on QQP overall (+18.2 pp) and SQuAD overall (+11.0 pp, driven by MFT accuracy); the AR family wins significantly on the surface-perturbation INV families: sentiment INV (−5.8 pp) and SQuAD INV (−6.8 pp). The coexistence of a diffusion win on SQuAD *overall* with an AR win on SQuAD *INV* is the task-conditional pattern, visible inside a single suite. At the model level, the three highest subtest-macro pass rates belong to diffusion models (LLaDA-8B 82.7%, Dream 81.5%, LLaDA-MoE 77.0%; best AR: Qwen2.5 75.7%).

Model	Type	Sentiment	QQP	SQuAD
LLaDA-8B	DLM	1.614	1.807	5.246
LLaDA-MoE	DLM	1.807	1.923	5.257
Dream-7B	DLM	1.948	1.540	5.226
Llama-3.1	AR	1.691	2.017	5.412
Mistral-7B	AR	1.811	2.411	5.625
Qwen2.5-7B	AR	1.630	0.808 [†]	5.032
Gemma-2-9B	AR	2.045	0.711 [†]	5.392
OLMo-2-7B	AR	1.629	2.071	4.904

Table 2: Per-model pair-weighted coefficients (α for DLMs, γ for ARs; pooled). Lower = more stable. [†]label-prior corruption, not robustness (§4.5); excluding the two in a diagnostic sensitivity analysis gives $\gamma_{\text{AR}} = 2.17$, $\gamma/\alpha = 1.23$ (1.48 INV-only).

4.2 Layer 2: relative perturbation coefficients

Table 2 gives the per-model pair-weighted coefficients and Figure 1 visualises the pairwise ratio $\gamma_{\text{AR}}/\alpha_{\text{Diff}}$ for every diffusion–AR pair. On QQP (the paraphrase suite) every diffusion model beats Llama, Mistral, and OLMo (ratios 1.05–1.57), whereas the comparisons against Qwen and Gemma fall below 1. Retaining all five AR models gives an overall family ratio of approximately $\gamma/\alpha = 0.91$, so the all-model comparison does not favour diffusion. The label-prior audit in §4.5 shows that Qwen and Gemma exhibit near-rigid prediction behaviour; excluding these two models as a diagnostic sensitivity analysis yields $\gamma/\alpha = 1.23$ over all types (1.48 INV-only). The single cleanest datapoint is the true (q, paraphrase(q)) subtest, where LLaDA’s $\alpha = 0.0187$ against $\gamma_{\text{AR}} = 9.81$ (the five-AR mean): the ratio reaches $\sim 525\times$, though the small paraphrase input distance inflates the absolute ratio; the direction is robust. Sentiment and SQuAD sit at parity ($\gamma/\alpha \approx 0.98$ –1.01 pairwise).

Aggregating subtest-level $\gamma - \alpha$ gaps by capability (Figure 2) sharpens the pattern: diffusion models are significantly more stable than ARs on *semantic* perturbations (negation) and significantly *less* on *surface* perturbations (NER, SRL, taxonomy, fairness). Coreference and robustness appear in the same capability view but remain within the bootstrap interval. The statistical scope matters here: the global pair-level gaps are directional but mostly within bootstrap noise, and only the capability-level structure reaches significance.

4.3 Selective stability: INV vs DIR

A model that passed every INV case by always predicting the same label would look robust yet

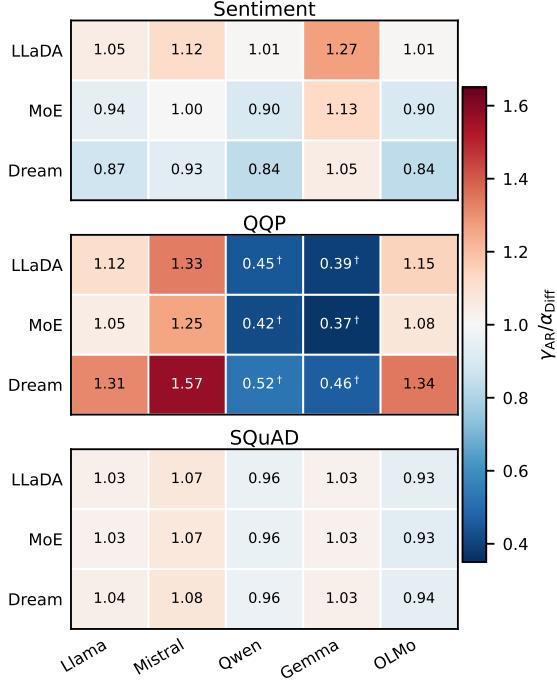


Figure 1: Pairwise perturbation-coefficient ratios by suite. Warm cells indicate $\gamma_{AR}/\alpha_{Diff} > 1$, where the DLM amplifies less than the paired AR baseline; cool cells indicate the reverse. Daggered QQP cells are the label-prior-corrupted Qwen/Gemma comparisons discussed in §4.5.

fail DIR completely. We confirm LLaDA’s stability is genuine, not degenerate: it passes 81.4% of QQP DIR cases, so it is not a constant-label model. The QQP DIR family splits cleanly along the semantic/entity boundary across seven subtests: LLaDA is near-perfect (95.5%) on the paraphrase-to-duplicate test (where Llama fails every single case) and under-moves only when a single changed entity flips the label. Every DIR subtest the AR wins is an entity-tracking test, so the aggregate DIR deficit is concentrated there, not spread across the family. Figure 3 maps the diffusion-versus-AR behavioural win share across all six test families. It separates the large SQuAD MFT gains from the sensitivity and surface-invariance regimes where stability costs (QQP DIR, sentiment INV, SQuAD INV), with QQP INV split at the subtest level.

4.4 Where the advantage reverses

When a perturbation *should* flip the answer (DIR), the comparison inverts. On QQP entity-swap tests (changing a name or number in one question of a duplicate pair), LLaDA fails 34–35% and LLaDA-MoE 58–60% of cases against 6–12% for the

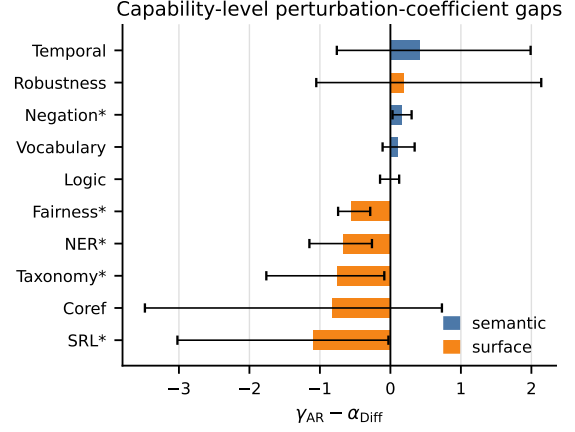


Figure 2: Mean $(\gamma_{AR} - \alpha_{Diff})$ by CheckList capability, pooled over the three suites. Positive = diffusion has the lower perturbation coefficient; * marks capabilities whose 95% CI excludes zero.

Llama reference. On SQuAD INV (fine-grained span localisation), LLaDA (82%) and LLaDA-MoE (76%) fall below the five-AR band (83–89%), though Dream (88%) sits inside it on every subtest, showing the SQuAD penalty is a LLaDA-family property, not an architectural inevitability. The diffusion models’ stability is useful for invariance tests but costly for sensitivity tests.

4.5 The QQP outlier is a label-prior artefact

The only sub-1 ratios are against Qwen and Gemma on QQP (Table 2); a label-distribution audit (Figure 4) shows these reflect a rigid label prior, not genuine robustness. Qwen answers `not_duplicate` on 98% of QQP, nominally the most “invariant” model (1.5% INV fail) yet wrong on 45% of MFT, and Gemma holds the mirror-image duplicate prior. A near-constant prediction forces the output shift $d(f(x'), f(x)) \rightarrow 0$ for every perturbation, which collapses γ_{AR} and spuriously makes the AR look “stable.” Qwen puts 54% of its QQP pair weight on subtests where the output never changes, so its pooled γ mostly reflects pairs that could not have moved in the first place. An option-order probe confirms the mechanism: reversing the instruction’s label order on 1,000 identical-question prompts flips 326/1000 of Qwen’s answers (a positional artefact), while the three diffusion models flip ≤ 1 ; and a label-free yes/no arm shows the deficit is not format-bound (Qwen still denies 690/1000 identical pairs). We therefore retain the all-model QQP comparison as the primary estimate and report exclusion of these two models as a diag-

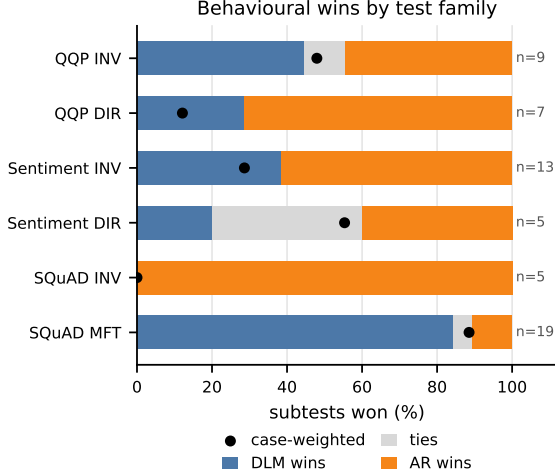


Figure 3: Selective stability: share of subtests won by the diffusion vs. AR family per test family, using family macro pass rates (bars), with case-weighted DLM win share shown as dots. Diffusion gains are concentrated in SQuAD MFT, while sensitivity (DIR) and surface-INV families favour ARs – the task-conditional pattern, not uniform robustness.

nostic sensitivity analysis. This shows how strongly the family-level coefficient comparison depends on label-prior rigidity.

5 Conclusions

In this study, we tested whether diffusion language models exhibit different robustness patterns from autoregressive models under CheckList perturbations, within the off-manifold framework of §2. Across eight 7–9B models and three behavioural suites, the result is mixed: which family wins depends on the task. Behaviourally, the diffusion family wins the overall QQP and SQuAD pass rates (+18.2 and +11.0 pp), but the AR family wins the sentiment and SQuAD surface-invariance slices. At the coefficient level, the all-model QQP comparison does not favour diffusion ($\bar{\gamma}_{AR}/\bar{\alpha}_{Diff} \approx 0.91$). Excluding two AR models with strong label-prior effects as a diagnostic sensitivity analysis changes the ratio to 1.23 overall and 1.48 INV-only. The advantage also reverses on perturbations that should change the answer, namely entity-swap and fine-grained span tests, so the diffusion models are not uniformly safer. Moreover, because the observed diffusion coefficients exceed one, the contraction assumption of Proposition 1 does not hold in this regime; the proposition should therefore be viewed as a conditional conceptual

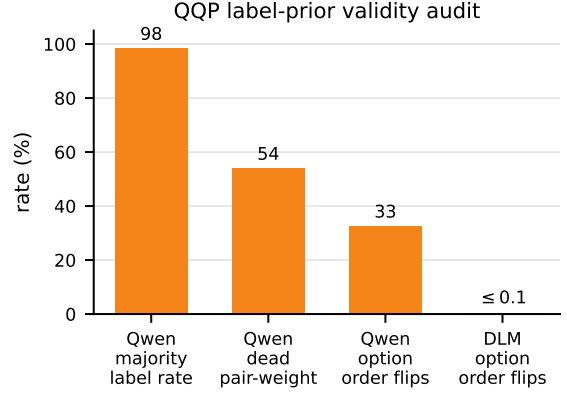


Figure 4: QQP label-prior audit. Qwen answers one label on 98% of pairs, carries 54% dead pair-weight, and flips 33% of answers under an option-order swap; the diffusion models flip $\leq 0.1\%$. The sub-1 QQP ratios are a rigid prior, not robustness.

framework rather than as directly validated by these experiments. Finally, the validity protocol (a decoding-budget floor and a label-distribution audit) matters: without it, under-denoising and AR label-prior rigidity fabricate or invert several cells.

Limitations

Our evaluation covers only the three original CheckList suites (sentiment, QQP, SQuAD), so the coefficient comparison speaks to those tests and not to behavioural testing in general. The comparison is also relative rather than absolute: both coefficients exceed one ($\alpha, \gamma > 1$), so diffusion amplifies a perturbation less than AR but never contracts it, and assumption (A3) ($\alpha < 1$) does not hold in this regime. Our models share a narrow setup, all eight in the 7–9B band, decoded deterministically with a fixed denoising budget, and we do not vary scale, sampling, or step count. Where the advantage appears it is task-conditional and reverses on the sensitivity (DIR) tests, so it is not architecture-wide robustness. Finally, most pair-level gaps sit within bootstrap noise, with only the capability- and family-level pass rates resolved, and we treat the off-manifold component (Assumption 1) as a modelling assumption for CheckList perturbations rather than measuring it per pair.

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and interpretation, are exclusively the work of the authors.

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Appendix A: Proof of Proposition 1

Proof. Condition on the event $T(x) \notin \mathcal{M}_\tau$, so that the perturbation has a non-trivial off-manifold component. By the scaled-loss assumption,

$$c d(u, v) \leq L(u, v) \leq C d(u, v), \quad u, v \in \mathcal{X}.$$

For the autoregressive model, the amplification assumption gives

$$d(f_{\text{AR}}(T(x)), f_{\text{AR}}(x)) \geq \gamma d(T(x), x),$$

and hence

$$L(f_{\text{AR}}(T(x)), f_{\text{AR}}(x)) \geq c\gamma d(T(x), x).$$

For the diffusion model, the contraction assumption gives

$$d(f_{\text{Diff}}(T(x)), f_{\text{Diff}}(x)) \leq \alpha d(T(x), x),$$

and hence

$$L(f_{\text{Diff}}(T(x)), f_{\text{Diff}}(x)) \leq C\alpha d(T(x), x).$$

Combining the two bounds,

$$\frac{L(f_{\text{Diff}}(T(x)), f_{\text{Diff}}(x))}{L(f_{\text{AR}}(T(x)), f_{\text{AR}}(x))} \leq \frac{C\alpha}{c\gamma}.$$

If $(C/c)(\alpha/\gamma) < 1$, then the diffusion loss is strictly smaller pointwise on the conditioned event. Taking expectations over $x \sim P_{\text{train}}$ and $T \sim \mathcal{D}$, conditioned on $T(x) \notin \mathcal{M}_\tau$, yields $\Delta_\perp(f_{\text{Diff}}) < \Delta_\perp(f_{\text{AR}})$. ■